

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(iii)		$K_c = \frac{[\text{H}^+][\{\text{Fe}(\text{H}_2\text{O})_5(\text{OH})\}^{2+}]}{[\{\text{Fe}(\text{H}_2\text{O})_6\}^{3+}]} \quad (1)$ unit $\Rightarrow$ mol dm ⁻³ (1)		1 1		2	1	
		(iv)		$[\text{H}^+] = 0.0282 \text{ mol dm}^{-3} \quad (1)$ $[\text{Fe}(\text{H}_2\text{O})_6]^{3+} = \frac{0.103 \times 0.0282}{4.03 \times 10^{-3}} \quad (1)$ $[\text{Fe}(\text{H}_2\text{O})_6]^{3+} = 0.720 \quad (1)$ mass of FeCl ₃ .6H ₂ O = (0.720 + 0.103) × 270.4 = 222.5g (1)		1	3	4	3	
	(e)	(i)		$[\text{Fe}(\text{H}_2\text{O})_3(\text{OH})_3] \text{ or } \text{Fe}(\text{OH})_3$ brown / red-brown / dark brown formula and colour needed	1			1		1
		(ii)		<u>oxygen</u> (from the air) can oxidise <u>Fe²⁺</u> to Fe ³⁺ (turning the precipitate brown) (1) award (1) for either of following - because oxygen has a more positive standard electrode potential than Fe³⁺ so it is a stronger oxidising agent - the EMF for the reaction between O₂ and Fe²⁺ is positive / +0.46V and positive reactions are feasible		1	1	2		1